Molecules 2010, 15 6822 Antioxidative Diet Diet and nutrition may affect the onset and course of chronic inflammatory airway diseases. Serum lycopene and vitamin A concentrations have been found to be significantly lower in asthmatics than in those without asthma [132,133]. In contrast, vitamin E intake is generally unrelated to asthma status but the level of vitamin E in serum is significantly lower in severe asthmatics than in mild asthmatics [133]. Among children, consumption of fresh fruits, particularly fruits high in vitamin C, has been related to a lower prevalence of asthma symptoms and improved lung functions [132,133]. Dietary supplementation or adequate intake of lycopene, vitamin A, flavonoids, and vitamin C rich foods may be beneficial in asthmatic subjects. As for the prevention of the airway disease, many foods may affect development of COPD [132]. Food rich in vitamin C and E may play an especially important role in the prevention and treatment of bronchial asthma. In fact, studies on lung function decrement and bronchial asthma in adults suggest that daily intake of vitamin C at levels exceeding the current recommended dietary allowance (60 mg/day among nonsmokers and 100 mg/day among smokers) may have a protective effect [132]. In one study, an increase of 40 mg/day in vitamin C intake led to an approximate 20 mL increase in FEV1 [132]. An epidemiological study on the impact of dietary selenium and carotenoids on major causes of mortality and morbidity (including asthma) in elderly women showed a lower risk of mortality in subjects with higher serum selenium and serum total carotenoids levels [175]. Volatile Compounds Phytoncides Phytoncides are natural volatile compounds emitted by trees and plants as a protective mechanism against the harmful insects and animals or microorganism. The major ingredients of phytoncide are highly volatile terpenoids such as -pinene, carene, and myrcene [176]. Terpenoids are frequently found in essential oils from plants. Monoterpenoids have been isolated from the oils of many higher plants and are valuable in the perfumery and flavor industries. They are also found in nature as insect pheromones and defense secretions and in many marine organisms, where they are usually halogenated. There are four different types of phytoncide solutions introduced by Japanese researchers; chemical components released from trees (A-type), highly bactericidal plants (AB-type), flowering grass (D- type), and non-allergic plants (G-type) [176]. Inhalation of phytoncides is reported as forest bathing and aromatherapy (relaxing effects) [177, 178] as well as antibacterial effects [179,180]. Chemical and pharmacological studies have shown that some species produce active principles that exert anti- gastropathic, anti-inflammatory, anti-oxidant, and radical scavenging activity [181-183]. It has been reported that physiological effects of phytoncides contribute to the improvement of various disorders including accelerated aging, allergies, multiple sclerosis, and Parkinson disease [176]. In addition, phytoncide is reported to enhance the activity of human NK cells [184]. It has also been introduced that fragrance from trees has a regulatory effect on immune function in humans and a restorative effect on the stress-induced immune suppression in mice [185,186]. There is no report about the roles of phytoncide in asthma. However, we have found that administration of phytoncide solution reduces asthmatic phenotypes of an animal model of asthma,